



Octave Programming Tutorial

As per Octave website (<http://www.octave.org/>), "Octave is a high-level language, primarily intended for numerical computations. It provides a convenient command line interface for solving linear and nonlinear problems numerically, and for performing other numerical experiments using a language that is mostly compatible with Matlab. It may also be used as a batch-oriented language."

The purpose of this collection of tutorials is to get you through most (and eventually all) of the available Octave functionality from a basic level.

A longer and more advanced Wikibook related to Octave Programming is the MATLAB Programming Wikibook. Those familiar with MATLAB that want to jump into Octave should consult MATLAB Programming/Differences between Octave and MATLAB.

Indeed, Octave is often viewed as a system for numerical computations with a language that is mostly compatible with Matlab, but that is available as free software under the GNU GPL, and that can replace it in many circumstances. This is why only one advanced programming Wikibook is being written; but therein the differences between MATLAB and Octave languages are presented.

Contents

Available tutorials for beginners:

- Getting started (complete)
- Vectors and matrices (complete)
- Plotting
- Text and file output (complete)
- General mathematical functions
- Loops and conditions
- Writing functions
- Vectorization

Available tutorials that are more advanced:

- Linear algebra (complete)
- Differential equations
- Polynomials (complete)
- Sets
- Filter design



A ***printable version*** of Octave Programming Tutorial is available. (edit it (https://en.wikibooks.org/w/index.php?title=Octave_Programming_Tutorial/Print_version&action=edit&preload=Template%3APrint+version%2FPreload))

Authors

- Henri Amuasi (updated by Carl Scheffler and Mike Pickles)

See also

- [commons:Category:Images with Octave source code](#)
- [INTLAB - INTerval LABoratory The Matlab/Octave toolbox for Reliable Computing by Siegfried M. Rump \(<http://www.ti3.tu-harburg.de/rump/intlab/>\)](#)

References

- [Octave \(<http://web.archive.org/web/20070607162216/http://www.aims.ac.za/wiki/index.php/Octave>\)](http://web.archive.org/web/20070607162216/http://www.aims.ac.za/wiki/index.php/Octave) A great deal of this tutorial has been copied from this location, that is GFDL.
- <http://www-h.eng.cam.ac.uk/help/programs/octave/tutorial/>

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